

EC 9540: Human Computer Interaction

LAB 03

Project Planning Report

Smart Hostel Maintenance Reporting System

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Contents

1. Introduction
2. Interface Design
 - 2.1 Login / Access Screen
 - 2.2 Home Screen
 - 2.3 Report Issue Screen
 - 2.4 My Complaints Screen
 - 2.5 Complaint Detail Screen
 - 2.6 Feedback Screen
3. Menus
4. GUI Elements
5. Navigation
6. Back-End Plan
 - 6.1 Data Requirements
 - 6.2 Database Requirements
 - 6.3 Major Back-End Functions
7. Overall System Model
8. Discussion
9. Conclusion

1. Introduction

This report presents the project planning stage that follows the Requirement Analysis and Hierarchical Task Analysis (HTA) documented in Lab 02 for the Smart Hostel Maintenance Reporting System. Lab 02 established that the primary target user the hostel student needs a fast, trackable, and low-effort way to report and follow up on maintenance issues, based on the analysis of 122 questionnaire responses.

Building directly on those findings and on the Goal-0/HTA breakdown (Sections 6-9 of Lab 02), this report plans the front-end and back-end design of the system across six areas: interface design, menu structure, GUI elements, navigation, back-end data/function plan, and the overall system model connecting the front end and back end.

Design decisions throughout this report are traceable to specific Lab 02 findings for example, the mobile-first, dark/light, multilingual, large-button, 1-2 step design directly reflects Sections 3.2 and 4.2 of the Lab 02 report.

2. Interface Design

The interface is planned as a mobile-first application (reflecting that 82.0% of respondents primarily use a mobile phone and 68.0% prefer a mobile app). Six core screens cover the full task cycle identified in the Lab 02 HTA: Access → Report → Track → Feedback. Each wireframe below follows the "Very Simple" layout preference (53.3% of respondents) with large, clearly labelled buttons (rated 4.52/5 in importance).

2.1 Login / Access Screen

Corresponds to HTA Sub-task 1 (Access the reporting system). Supports hostel ID login or continuing an authenticated session, plus language and theme selection up front.

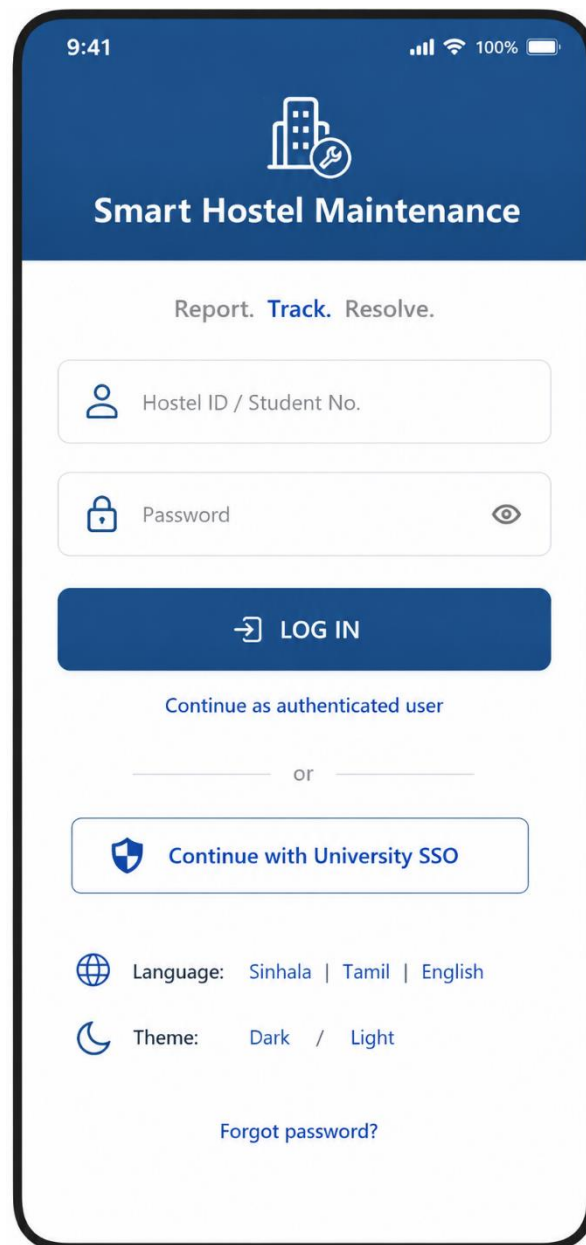


Figure 2.1 - Login / Access screen wireframe

2.2 Home Screen

The landing screen after login. Surfaces the emergency-report shortcut (a top-requested feature, 40.2%) and the primary "Report Issue" action, plus a preview of active complaints so students see status without extra navigation.

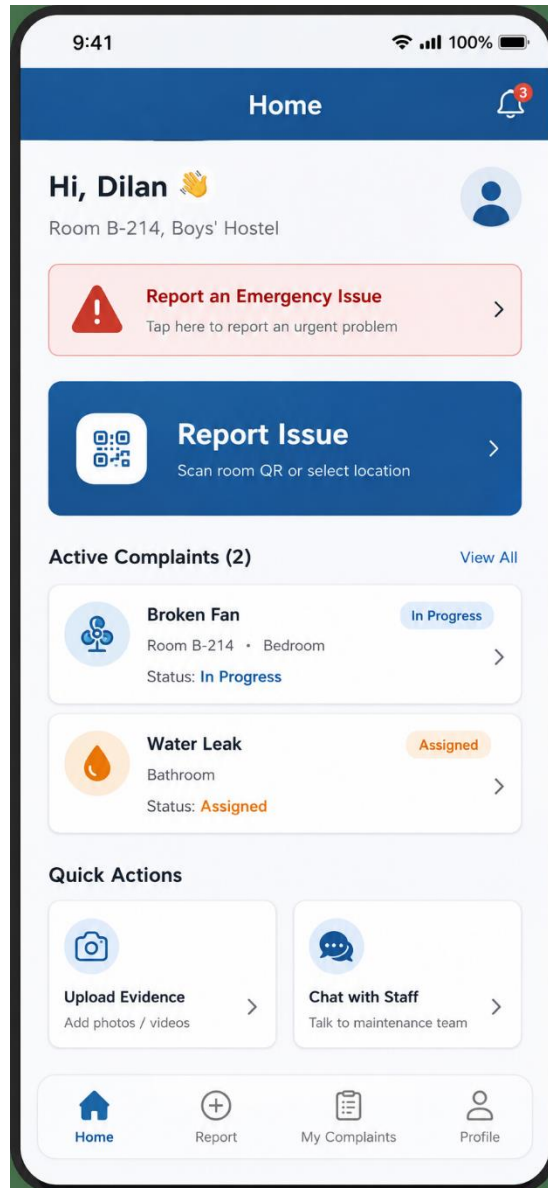
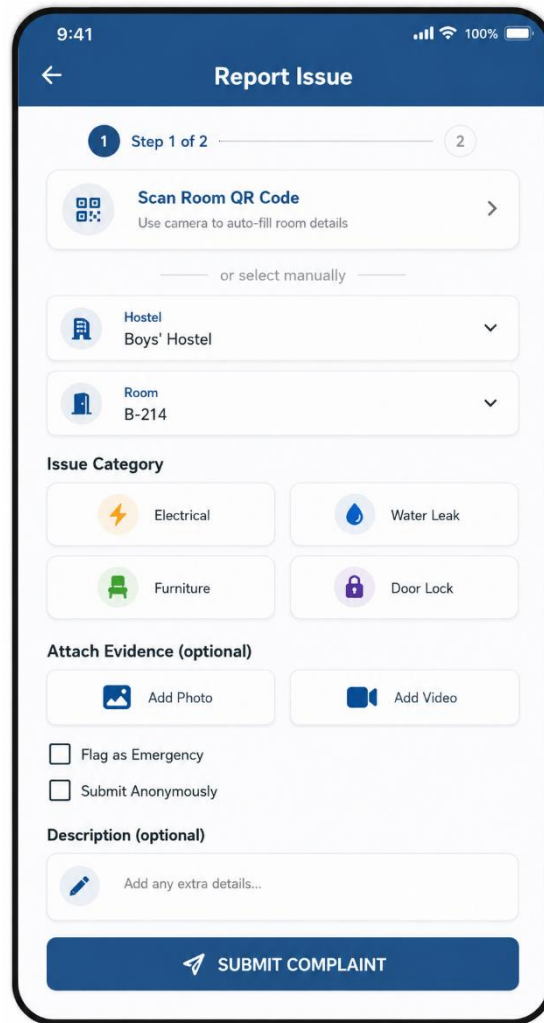


Figure 2.2 - Home screen wireframe

2.3 Report Issue Screen

Implements HTA Sub-task 2. Designed to be completed in 1-2 steps (85.2% preference): QR scan or manual location selection, category selection, optional photo/video, emergency/anonymous flags, and a single submit action.



The wireframe shows a mobile app screen titled "Report Issue". At the top, there's a status bar with the time 9:41, signal strength, Wi-Fi, and 100% battery. Below the title bar, a progress indicator shows "Step 1 of 2". The main content area starts with a "Scan Room QR Code" button, which includes a QR icon and the text "Use camera to auto-fill room details". Below this is a link "or select manually". Then, there are two dropdown menus: "Hostel" (selected: Boys' Hostel) and "Room" (selected: B-214). The "Issue Category" section has four buttons: "Electrical" (lightning bolt icon), "Water Leak" (water drop icon), "Furniture" (chair icon), and "Door Lock" (lock icon). The "Attach Evidence (optional)" section has two buttons: "Add Photo" (photo icon) and "Add Video" (video icon). Below these are two checkboxes: "Flag as Emergency" and "Submit Anonymously". The "Description (optional)" section has a text input field with a pencil icon and the placeholder text "Add any extra details...". At the bottom is a large blue button with a paper plane icon and the text "SUBMIT COMPLAINT".

Figure 2.3 - Report Issue screen wireframe

2.4 My Complaints Screen

Implements the start of HTA Sub-task 3. Filterable by All / Active / Resolved, with color coded status dots addressing the top pain point that 50.0% of students never know a complaint's status.

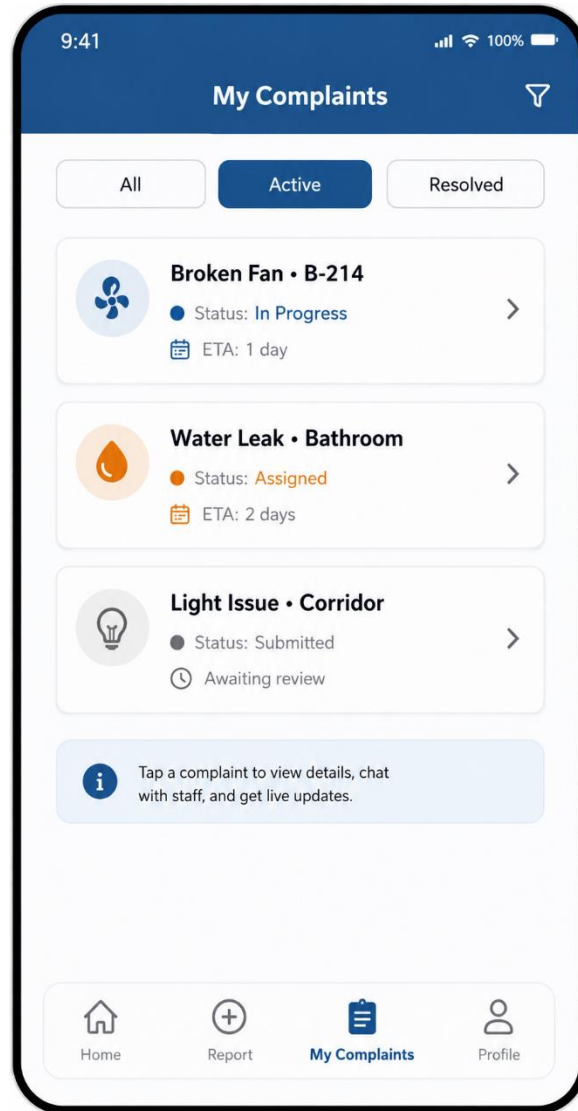


Figure 2.4 - My Complaints (tracking list) screen wireframe

2.5 Complaint Detail Screen (Status Tracker + Chat)

Continues HTA Sub-task 3. A horizontal progress tracker shows the four-stage status (Submitted, Assigned, In Progress, Resolved), alongside an in-app chat with the assigned maintenance staff the second most-requested feature (33.6%).

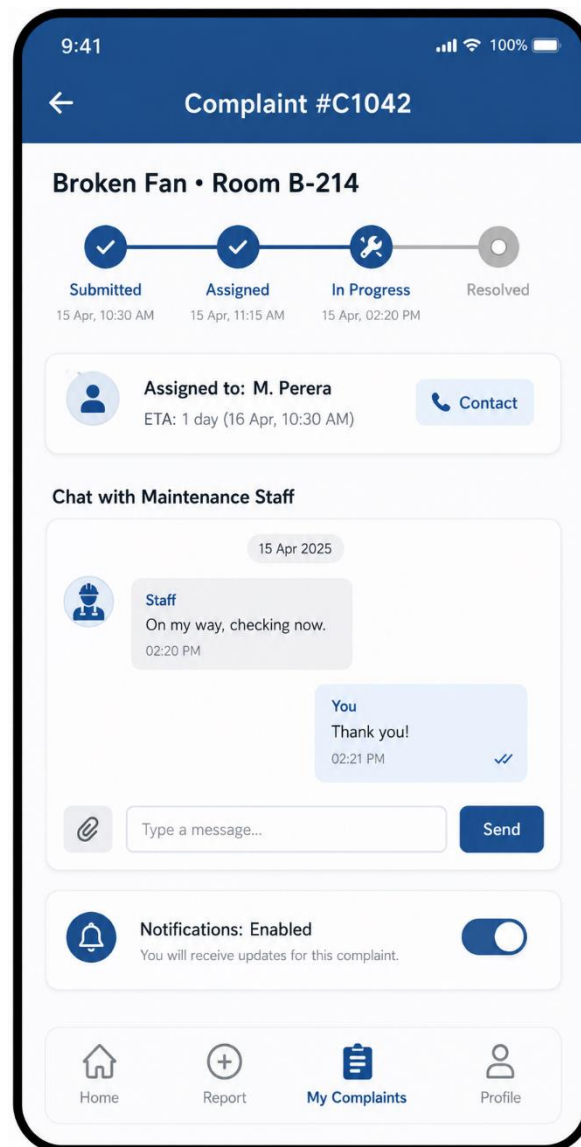


Figure 2.5 - Complaint Detail screen wireframe with status tracker and chat

2.6 Feedback Screen

Implements HTA Subtask 4. Appears once a complaint reaches Resolved status; captures a staff rating and a repair-speed rating plus optional comments, then closes the complaint into history.

9:41 100%

← Repair Feedback

✓

Issue Resolved!

Broken Fan • Room B-214

Rate the Maintenance Staff

★ ★ ★ ★ ☆

Rate the Speed of Repair

★ ★ ★ ★ ★

Additional Comments (optional)

Describe the outcome...

SUBMIT FEEDBACK

This closes the complaint and adds it to your complaint history.

Figure 2.6 - Feedback and rating screen wireframe

3. Menus

The system uses a persistent bottom navigation bar as the primary menu on the student mobile app (matching the mobile-first, low-step design goal), supplemented by contextual menus on individual screens. Secondary users (Warden, Sub-Warden, Maintenance Staff) access an administrative web dashboard with its own menu, since Lab 02 identified these as secondary/operational users of the system.

Menu / Item	Description
Bottom Navigation Bar (Student App, persistent)	
• Home	Dashboard, emergency shortcut, Report Issue entry point, active complaints preview
• Report	Opens the Report Issue form directly (Sub-task 2)
• My Complaints	Tabs: All / Active / Resolved — Sub-task 3
• Profile	Account details, language, theme, log out
Home Screen Contextual Menu	
• Report an Emergency Issue	Priority shortcut, pre-flags the complaint as emergency
• Upload Evidence	Quick action shortcut into the Report Issue form
• Chat with Staff	Quick action into the most recent active complaint's chat
Complaint Detail Menu	
• Status Tracker	Read-only visual progress (Submitted → Resolved)
• Chat	In-app messaging with assigned maintenance staff
• Feedback (appears once Resolved)	Leads into the Feedback & Rating screen
Admin Web Dashboard Menu (secondary users)	
• Complaint Queue	All incoming complaints, filterable by hostel/status/urgency
• Staff Assignment	Assign maintenance staff to a complaint
• Analytics	Recurring-issue trends per room/hostel

4. GUI Elements

The table below itemises the concrete GUI elements buttons, fields, labels, controls planned for each screen, mapped to the wireframes in Section 2 and to the functional requirements (FR) identified in Lab 02, Section 4.1.

Login / Access

Element Type	Element / Label	Purpose
Text field	Hostel ID / Student No.	Captures student identifier
Password field	Password	Masked credential entry
Primary button	LOG IN	Submits credentials
Link / label	Continue with University SSO	Alternate authentication
Dropdown / toggle	Language selector (Sinhala / Tamil / English)	Sets app language
Toggle	Theme (Dark / Light)	Sets colour theme
Link	Forgot password?	Password recovery

Home

Element Type	Element / Label	Purpose
Label	Greeting + room/hostel	Personalises the screen
Banner button	Report an Emergency Issue	Priority shortcut for urgent issues
Primary button (large)	Report Issue	Entry point to Sub-task 2
List / cards	Active Complaints preview	Quick status glance
Secondary buttons	Upload Evidence, Chat with Staff	Quick actions
Bottom navigation bar	Home / Report / My Complaints / Profile	Persistent app-wide navigation

Report Issue

Element Type	Element / Label	Purpose
Button	Scan Room QR Code	Auto-fills location (FR4)
Dropdown	Hostel, Room (manual fallback)	Manual location selection
Selectable chips	Issue category (Electrical, Water Leak, Furniture, Door Lock)	Category selection (FR1)
File upload buttons	Add Photo, Add Video	Optional evidence attachment (FR1)
Checkbox	Flag as Emergency	Priority handling (FR5)
Checkbox	Submit Anonymously	Anonymous reporting (FR6)

Element Type	Element / Label	Purpose
Text area	Description (optional)	Free-text details
Primary button	Submit Complaint	Finalises Sub-task 2

My Complaints

Element Type	Element / Label	Purpose
Tab control	All / Active / Resolved	Filters the complaint list
List / cards	Complaint summary (title, status dot, ETA)	Complaint tracking (FR2)
Status indicator	Colour-coded status dot	Visual status cue
Bottom navigation bar	shared component	App-wide navigation

Complaint Detail

Element Type	Element / Label	Purpose
Stepper / progress tracker	Submitted → Assigned → In Progress → Resolved	Visualises real-time status (FR2)
Label	Assigned staff + ETA	Repair time estimate (FR7)
Chat window	Message bubbles (staff / student)	In-app chat (FR10)
Text input + Send button	Type a message...	Compose chat message
Status label	Notifications: Enabled	Push notification confirmation (FR3)

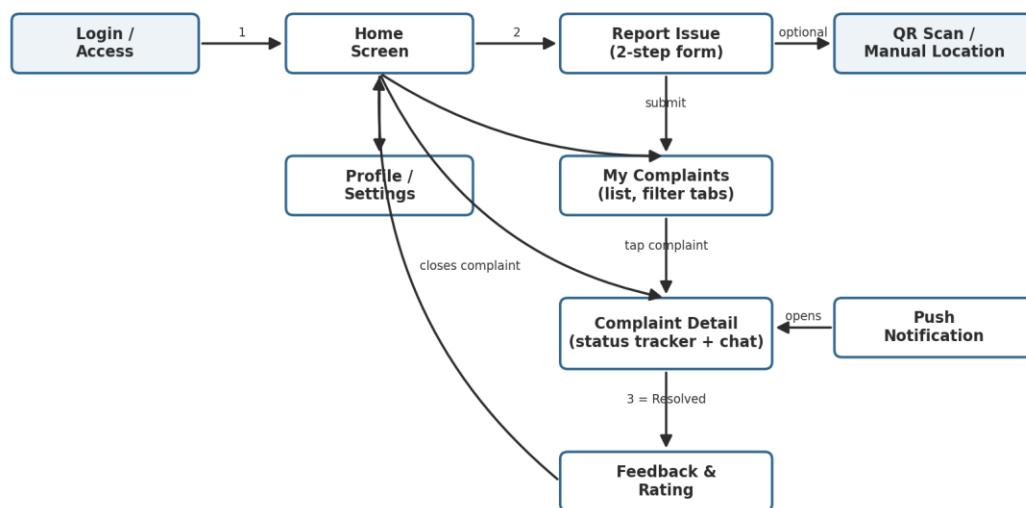
Feedback

Element Type	Element / Label	Purpose
Label / icon	Issue Resolved confirmation	Confirms repair completion
Star rating control	Rate the Maintenance Staff	Staff rating (FR8)
Star rating control	Rate the Speed of Repair	Speed rating (FR8)
Text area	Additional Comments (optional)	Free-text feedback
Primary button	Submit Feedback	Closes complaint, updates history (FR9)

5. Navigation

Navigation follows the sequential flow of the HTA in Lab 02 (Plan 0: do 1-2-3, then 4 once resolved), while the bottom navigation bar gives direct shortcuts between the four main sections at any time. Push notifications provide a secondary entry point directly into a complaint's detail screen when its status changes.

Navigation Flow — Smart Hostel Maintenance Reporting System (Student App)



Bottom Navigation (persistent): Home · Report · My Complaints · Profile

Numbered arrows show the primary task sequence (Sub-tasks 1-4, Section 7/8 of Lab 02).
Curved arrows show shortcuts available from the Home screen via the bottom navigation bar.

Figure 5.1 - Navigation flow across the student app

6. Back-End Plan

6.1 Data Requirements

The back end must persist student and staff accounts, complaints and their status history, media attachments, chat messages, and feedback. The simplified entity overview below shows the core tables and their relationships.

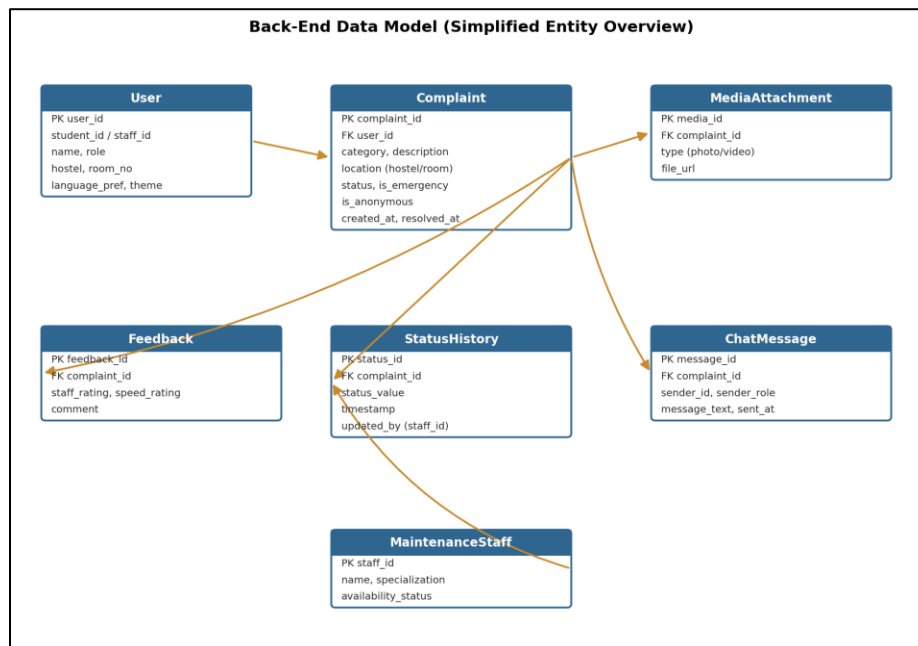


Figure 6.1 - Simplified back-end data model

6.2 Database Requirements

- A relational database (e.g., PostgreSQL / MySQL) for structured entities: User, Complaint, StatusHistory, Feedback, MaintenanceStaff, ChatMessage.
- Cloud object storage (e.g., S3-compatible bucket) for photo/video evidence, referenced by URL from the Complaint/MediaAttachment table.
- A lightweight notification queue integrated with a push service (e.g., Firebase Cloud Messaging / Apple Push Notification service).
- Indexing on complaint status, user_id, and room/hostel fields to support fast status lookups and per-room complaint history (FR9).

6.3 Major Back-End Functions

Function	Description	Owning Service
Authentication	Register / log in a student or staff member; issue session tokens	User & Auth Service
Complaint creation	Validate and store a new complaint, incl. location, category, media, flags	Complaint Service
Status update	Allow assigned staff to change complaint status and record history	Complaint Service
Complaint retrieval	Return a student's active/resolved complaints and complaint detail	Complaint Service
QR resolution	Decode a scanned room QR code into hostel/block/room identifiers	Complaint Service
Notification dispatch	Send a push notification whenever complaint status changes	Notification Service
Chat messaging	Store and deliver real-time messages between student and staff	Chat Service
Feedback capture	Store staff/speed ratings and comments; close the complaint	Complaint Service
Complaint history	Aggregate past complaints per student and per room for trend analysis	Analytics Module
Media handling	Upload, store, and serve photo/video evidence via cloud storage	Media Storage

7. Overall System Model

The overall system connects the student mobile app and the administrative web interfaces to a set of back-end services through a single API gateway, which in turn read/write the data and storage layer. This model directly supports every functional requirement from Lab 02 Section 4.1, including real-time status tracking, push notifications, QR-based room identification, in-app chat, and post-repair feedback.

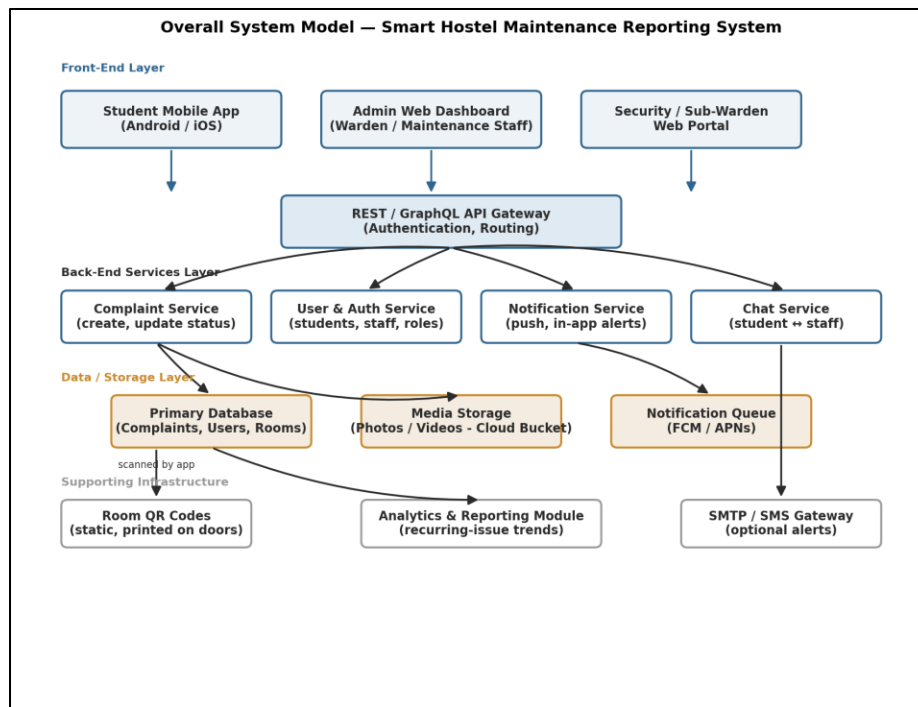


Figure 7.1 - Overall system architecture connecting front end and back end

Front-end layer: the Student Mobile App is the primary client (matching the 68.0% mobile-app preference), while the Admin Web Dashboard and Security/Sub-Warden Web Portal serve secondary/administrative users identified in Lab 02.

Back-end services layer: four focused services Complaint, User & Auth, Notification, and Chat sit behind a single API gateway that handles authentication and routing, keeping each service independently scalable.

Data/storage layer: a primary relational database stores structured records, cloud media storage holds photo/video evidence, and a notification queue manages reliable push delivery.

Supporting infrastructure: static room QR codes are scanned by the app to auto-fill location data, an analytics module surfaces recurring-issue trends from complaint history, and an optional SMTP/SMS gateway extends notifications beyond push alerts.

8. Conclusion

This project planning report presents the proposed design of the Smart Hostel Maintenance Reporting System based on the user requirements and Hierarchical Task Analysis identified in Lab 02. The planning process covered the major aspects required for system development, including interface design, menu structure, GUI elements, navigation flow, back-end requirements, database planning, and the overall connection between the front-end and back-end components.

The proposed system follows a mobile-first and user-centred design approach because the Lab 02 findings showed that students prefer a simple and efficient method for reporting maintenance issues. Features such as

QR-based room identification, emergency reporting, media attachment, complaint tracking, push notifications, in-app communication, and feedback were included to address the main problems and expectations identified during requirement gathering.

The navigation and GUI design aim to minimize user effort while providing clear system feedback throughout the maintenance-reporting process. At the same time, the proposed back-end architecture provides the necessary support for complaint management, authentication, status updates, notifications, communication, media handling, and complaint history.

Therefore, this planning report provides a clear foundation for the next stages of the project. The proposed interfaces and system structure can be further evaluated through prototype testing and user feedback before moving towards detailed implementation.

9. Discussion

The proposed design of the Smart Hostel Maintenance Reporting System was developed based on the user requirements and task analysis identified in Lab 02. The questionnaire findings showed that students require a simple, fast, mobile-friendly method for reporting hostel maintenance problems and tracking their progress. Therefore, the proposed interfaces focus on reducing the number of steps required to submit a complaint while providing clear feedback about complaint status.

The mobile-first design, large buttons, simple navigation, QR-based room identification, complaint tracking, push notifications, and in-app communication were selected to reduce the effort required from students. These features also address important problems identified during the requirement analysis, particularly the difficulty students experience in knowing whether a reported issue has been received, assigned, or resolved.

The planned navigation structure allows users to access the main functions through a persistent bottom navigation bar. This reduces unnecessary navigation and helps users quickly reach commonly used functions such as reporting an issue and checking existing complaints. The use of status indicators and a progress tracker also improves system visibility by clearly communicating the current state of each maintenance request.

The back-end design supports the front-end requirements by maintaining user information, complaint records, status history, media attachments, chat messages, and feedback. Separating major functions such as authentication, complaint management, notification handling, and messaging also provides a clear structure for future system development.

Overall, the proposed design attempts to balance usability, functionality, and technical feasibility. However, the current interfaces are planning-level wireframes and should be evaluated with representative hostel students before final implementation. User feedback obtained during usability testing can then be used to improve navigation, terminology, interface layout, accessibility, and overall interaction efficiency.

10. References

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